

Q3a) List and Elaborate Any Five JSP Implicit Objects with Example

JSP Implicit Objects - JSP provides built-in objects called **implicit objects**. These objects are automatically available in JSP pages without explicit declaration. There are total 9 implicit objects in JSP.

1) request Object - request object is used to get data sent by client request.

Common Methods - `getParameter()`, `getAttribute()`

Example - `<%`

```
String name = request.getParameter("uname");
out.println(name);
%>
```

2) response Object - response object is used to send response to the client.

Example - `<%`

```
response.sendRedirect("home.jsp");
%>
```

3) out Object - out object is used to display output on browser. It is an object of `JspWriter`.

Example - `<%`

```
out.println("Welcome to JSP");
%>
```

4) session Object - session object is used to store user-specific data across multiple requests.

Example - `<%`

```
session.setAttribute("user", "Rahul");
String s = (String)session.getAttribute("user");
out.println(s);
%>
```

5) application Object - application object is used to share data among all users of the application.

Example -

```
<%
application.setAttribute("college", "SPPU");
out.println(
application.getAttribute("college"));
%>
```

Other JSP Implicit Objects - `config`, `page`, `pageContext`, `exception`,

Advantages of JSP Implicit Objects - Easy access to request/response data, Reduces coding effort, Simplifies web application development.

JSP implicit objects help developers access web resources easily without creating objects manually, making JSP development simpler and faster.

Q3b) Explain the Struts Architecture with Neat Diagram and Benefits of Struts

Struts Framework - Struts is an open-source Java framework used for developing web applications based on the MVC (Model View Controller) architecture.

It simplifies enterprise web application development.

Components of Struts Architecture - Model, View & Controller

Struts MVC Architecture -

1) Model

- Represents business logic and database operations
- Usually implemented using JavaBeans or EJB

2) View

- Represents presentation layer
- Developed using JSP pages

3) Controller

- Handles client requests
- Implemented using ActionServlet

Struts Architecture Diagram -

Working of Struts Framework

1. User sends request
2. Request goes to ActionServlet
3. ActionServlet checks configuration file
4. Corresponding Action class is executed
5. Business logic is processed
6. Result sent to JSP page
7. Response displayed to user

Important Components of Struts

Component	Purpose
ActionServlet	Main controller
Action Class	Handles request processing
ActionForm	Stores form data
JSP	Presentation layer
struts-config.xml	Configuration file

Benefits of Struts

- 1) **MVC Architecture Support** - Separates business logic, presentation, and control logic.
- 2) **Reusable Components** - Code reuse becomes easier.
- 3) **Easy Maintenance** - Applications are easier to maintain and modify.
- 4) **Faster Development** - Provides built-in libraries and tags.
- 5) **Better Security** - Centralized request handling improves security.
- 6) **Form Validation Support** - Supports input validation.

Advantages of Struts - Platform independent, Open source, Scalable framework, Reduces development time

Struts framework simplifies Java web application development using MVC architecture. It improves maintainability, scalability, and code organization in enterprise applications.

Q4a) Write a JSP Program to Demonstrate Use of Page Directive, Scriptlet, Expression and Comment

JSP Elements - The following JSP elements are demonstrated: Page Directive, Scriptlet, Expression, Comment

JSP Program -

```
<%@ page language="java"
contentType="text/html"
%>
<html>
<head>
<title>JSP Demo</title>
</head>
```

```

<body>
<!-- HTML Comment -->
<%-- JSP Comment --%>
<h2>Demonstration of JSP Elements</h2>
<%
    // Scriptlet Example

    String name = "Rahul";
    int a = 10;
    int b = 20;
    int sum = a + b;
%>
Name is :
<%= name %>

<br><br>

Sum is :
<%= sum %>

</body>
</html>

```

Output - Demonstration of JSP Elements

Name is : Rahul

Sum is : 30

Explanation of JSP Elements -

1) Page Directive - Used to provide information to JSP container.

Syntax - <%@ page attribute="value" %>

Example Used - <%@ page language="java"
contentType="text/html" %>

2) Scriptlet - Used to write Java code inside JSP.

Syntax - <% Java Code %>

Example <% int sum = a + b; %>

3) Expression - Used to display output directly on browser.

Syntax - <%= expression %>

Example - `<%= sum %>`

4) Comment

JSP Comment - Not visible in browser source.

Syntax - `<%-- JSP Comment --%>`

HTML Comment - Visible in page source.

Syntax - `<!-- HTML Comment -->`

JSP provides various elements such as directives, scriptlets, expressions, and comments that simplify dynamic web page development.

Q4b) Identify and Justify the Benefits of Web Services

Web Services are software components used for communication between applications over the internet using standard protocols. They enable interaction between different platforms and languages.

Features of Web Services - Platform independent, Language independent, Interoperable, Reusable, Supports distributed computing

Benefits of Web Services

- 1) Interoperability** - Different applications developed in different languages can communicate easily. Example: Java application communicating with .NET application
- 2) Platform Independence** - Web services work on any operating system and platform.
- 3) Reusability** - Existing business logic can be reused in multiple applications.
- 4) Easy Integration** - Applications can be integrated easily using standard protocols like: HTTP, SOAP, XML
- 5) Cost Effective** - Reduces development and maintenance cost through reusable services.
- 6) Scalability** - Web services can handle growing business requirements efficiently.
- 7) Loose Coupling** - Client and server applications remain independent of each other.
- 8) Support for Distributed Applications** - Supports communication among distributed systems over networks.
- 9) Standardized Communication** - Uses standard technologies: - XML, SOAP, WSDL, UDDI

Advantages of Web Services - Improved business efficiency, Better communication between systems, Flexible and extensible.

Q3a) Explain JavaBeans Classes & JSP

JavaBeans are reusable software components written in Java that follow specific conventions.

They are used to encapsulate multiple objects into a single object (bean).

JavaBeans are mainly used in JSP for separating business logic from presentation logic.

Features of JavaBeans - Reusable components, Platform independent, Easy maintenance, Supports encapsulation

Rules of JavaBeans Class - A JavaBean class must:

1. Implement Serializable interface
2. Have default constructor
3. Provide getter and setter methods
4. Keep properties private

Example of JavaBean Class -

```
import java.io.*;

public class StudentBean implements Serializable {
    private String name;
    public void setName(String name) {
        this.name = name;
    }
    public String getName() {
        return name;
    }
}
```

Using JavaBean in JSP

```
<jsp:useBean id="s"
class="StudentBean" />
```

```
<jsp:setProperty
name="s"
property="name"
value="Rahul"/>
```

Name :

```
<jsp:getProperty
```

```
name="s"
property="name"/>
```

Advantages of JavaBeans - Code reusability, Simplifies JSP development, Separates business logic, Easy maintenance

JSP (Java Server Pages) is a server-side technology used to create dynamic web pages using HTML and Java.

JSP pages are converted into servlets by the web container.

Features of JSP - Dynamic content generation, Easy integration with JavaBeans, Platform independent, Simplifies web development

Simple JSP Example -

```
<html>
<body>
<%
    out.println("Welcome to JSP");
%>
</body>
</html>
```

Advantages of JSP - Easy to develop, Faster development, Supports reusable components, Reduces Java code complexity.

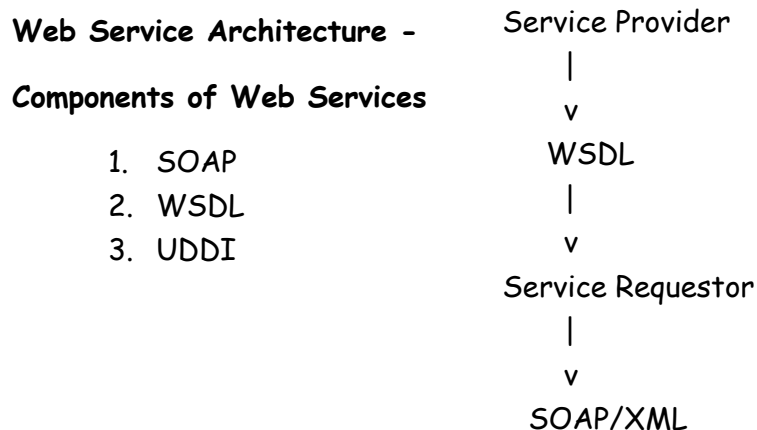
JavaBeans and JSP together help develop modular, reusable, and dynamic Java web applications efficiently.

Q3b) Explain the Concept of Web Services. Explain WSDL & SOAP.

Web Services are software components that allow communication between applications over the internet using standard web protocols.

They enable interaction between applications developed using different languages and platforms.

Features of Web Services - Platform independent, Language independent, Interoperable, Reusable, Distributed computing support



SOAP stands for: Simple Object Access Protocol

SOAP is an XML-based protocol used for exchanging information between applications over the internet.

Features of SOAP - XML based, Platform independent, Works over HTTP, SMTP, etc.

, Supports distributed systems

Structure of SOAP Message

1. Envelope
2. Header
3. Body
4. Fault

SOAP Message Example -

```
<soap:Envelope
xmlns:soap=
"http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <getEmployee>
      <id>101</id>
    </getEmployee>
  </soap:Body>
</soap:Envelope>
```

Advantages of SOAP - Secure communication, Platform independent, Standard protocol

WSDL stands for: Web Services Description Language

WSDL is an XML-based language used to describe web services.

It specifies: Service location, Methods provided, Parameters, Data types

Main Elements of WSDL

1. Types
2. Message
3. PortType
4. Binding
5. Service

WSDL Example -


```

<definitions>
  <message name="getRequest">
  </message>
  <portType name="EmployeePort">
  </portType>
  <service name="EmployeeService">
  </service>
</definitions>

```

Advantages of WSDL - Describes web services clearly, Supports interoperability, Easy service integration

Web services allow communication between heterogeneous applications. SOAP handles message exchange, while WSDL describes the functionality and structure of web services.

Q4a) Explain JSP Vs Servlets (Min. 04 Differences). Explain How JSPs are Translated to Servlets.

Basis	JSP	Servlet
Full Form	Java Server Pages	Java Servlet
Purpose	Used for presentation layer	Used for business logic
Coding Style	HTML with Java code	Pure Java code
Development	Easier to develop web pages	More difficult for UI design
Compilation	Converted into servlet internally	Already a Java class
Performance	Slightly slower initially	Faster execution
Maintenance	Easier for UI changes	Difficult for UI maintenance
Implicit Objects	Available automatically	Need explicit creation
Suitable For	Dynamic webpage design	Request processing
Extension	.jsp	.java

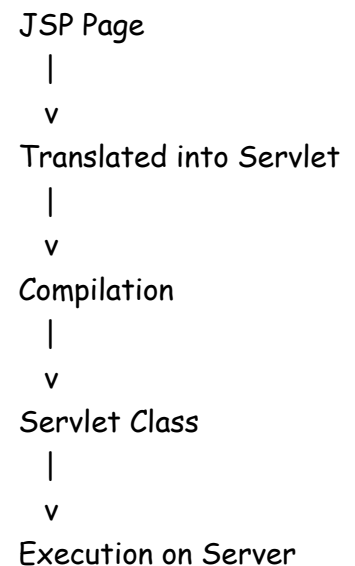
How JSPs are Translated to Servlets - A JSP page is internally converted into a servlet by the web container.

Steps of JSP Translation -

1. JSP page is written by developer

2. Web container translates JSP into servlet source code
3. Servlet source code is compiled into .class file
4. Servlet class is loaded into memory
5. `jspInit()` method is called
6. `_jspService()` method handles requests
7. `jspDestroy()` method destroys JSP

JSP Translation Diagram -



JSP simplifies webpage development, whereas servlets are mainly used for processing requests and business logic. JSPs are internally translated into servlets by the web container before execution.

Q4b) Write Note on Struts. Explain Struts with Example Code Block.

Struts ...

Component Purpose

Model	Business logic
View	JSP pages
Controller	ActionServlet

Example of Struts

1) JSP Page

```

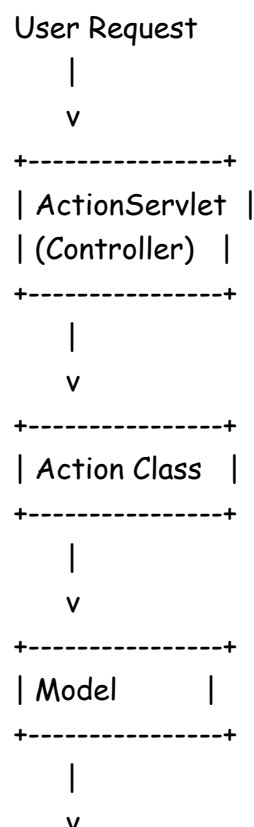
<html>
<body>
<form action="login.do">
Username:
<input type="text" name="uname">
<input type="submit" value="Login">
</form>
</body>
</html>
  
```

2) Action Class

```

import org.apache.struts.action.*;
public class LoginAction extends Action
{
  
```

Struts Architecture Diagram -



```

    public ActionForward execute( ActionMapping mapping, ActionForm form,
    HttpServletRequest req, HttpServletResponse res)  {
        return mapping.findForward("success");
    }
}

```

3) struts-config.xml

```

<action-mappings>
  <action
    path="/login"
    type="LoginAction">
    <forward
      name="success"
      path="/success.jsp"/>
    </action>
</action-mappings>

```

Struts framework simplifies Java web application development using MVC architecture and provides better maintainability, scalability, and code reusability.

Q3a) Explain the JSP Support for MVC Paradigm

MVC Paradigm - MVC stands for: Model - View - Controller

MVC is a software design pattern used for developing web applications by separating: Business logic, User interface, Request handling

Components of MVC -

Component Purpose

Model Handles business logic and database operations

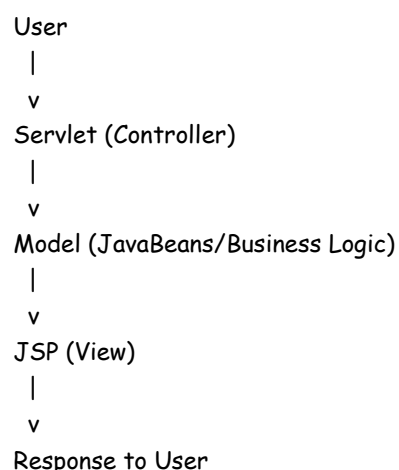
View Displays data to user

Controller Handles user requests and controls flow

JSP Support for MVC - In Java web applications:

- JSP acts as **View**
- Servlet acts as **Controller**
- JavaBeans/Classes act as **Model**

MVC Architecture using JSP -



Working of JSP in MVC

1. User sends request
2. Servlet receives request (Controller)
3. Servlet interacts with Model
4. Model processes business logic/database
5. Result sent to JSP
6. JSP displays response to user

Advantages of MVC using JSP - Separation of concerns, Easy maintenance, Better code reusability, Simplifies debugging, Scalable applications

JSP supports MVC architecture by acting as the presentation layer, while servlets and JavaBeans manage control and business logic respectively.

Q3b) Explain Struts Framework with Respect to Architecture, Actions, Interceptors & Exception Handling

Struts Framework ...

Components of Struts ...

Actions in Struts - Action classes process user requests and execute business logic.

Every action class extends: `ActionSupport`

Example of Action Class

```
public class LoginAction extends ActionSupport {  
    public String execute() {  
        return "success";  
    }  
}
```

Interceptors in Struts - Interceptors are objects used to intercept requests and responses before or after action execution.

They provide: Validation, Logging, Authentication, Exception handling

Working of Interceptors -

Common Interceptors -

- Validation Interceptor
- Params Interceptor

Request
|
Interceptor
|
Action
|
Result

- Exception Interceptor

Exception Handling in Struts - Struts provides centralized exception handling.

Exceptions can be handled globally using configuration files.

Example -

```
<global-exception-mappings>
  <exception-mapping
    exception="java.lang.Exception"
    result="error"/>
</global-exception-mappings>
```

Advantages of Struts - MVC support, Reusable code, Easy maintenance, Validation support, Better exception handling

Struts framework simplifies enterprise application development by providing MVC architecture, action classes, interceptors, and centralized exception handling.

Q4a) Explain JSP Lifecycle

JSP Lifecycle describes how a JSP page is processed by the web container from creation to destruction.

Phases of JSP Lifecycle

1. Translation Phase

2. Compilation Phase

3. Class Loading

4. Instantiation

5. Initialization

6. Request Processing

7. Destruction

JSP Lifecycle Diagram -

Lifecycle Methods

1) jspInit()

- Called once
- Used for initialization

JSP Page

|

v

Translation into Servlet

|

v

Compilation

|

v

Class Loading

|

v

jspInit()

|

v

_jspService()

|

v

jspDestroy()

```
public void jspInit() {
    System.out.println("JSP Initialized");
}
```

2) `_jspService()`

- Called for every request
- Generates response

```
public void _jspService( HttpServletRequest req, HttpServletResponse res) {
}
```

3) `jspDestroy()`

- Called once before JSP removal
- Releases resources

```
public void jspDestroy() {
    System.out.println("JSP Destroyed");
}
```

Advantages of JSP - Easy web page development, Dynamic content generation, Supports reusable components, Platform independent

JSP lifecycle manages the translation, compilation, execution, and destruction of JSP pages efficiently within the web container.

Q3a) List & Describe Important Interceptors Provided by Struts Framework

Interceptors in Struts - Interceptors are objects that intercept requests and responses before or after action execution in Struts framework.

They provide additional services such as: Validation, Logging, File upload, Exception handling, Parameter processing, Interceptors are part of Struts 2 framework.

1) Params Interceptor - Copies request parameters into action properties.

Example - private String username;

```
public void setUsername(String username) {
    this.username = username;
}
```

2) Validation Interceptor - Validates user input before action execution.

Example - `<field-validator type="requiredstring">`
`</field-validator>`

3) Exception Interceptor - Handles exceptions centrally.

Example -

```
<global-exception-mappings>
  <exception-mapping
    exception="java.lang.Exception"
    result="error"/>
</global-exception-mappings>
```

4) FileUpload Interceptor - Used for uploading files in web applications.

Features - File type validation, File size checking

5) Token Interceptor - Prevents duplicate form submissions.

Use - Useful in: Online payments, Registration forms

Advantages of Interceptors - Code reusability, Modular programming, Centralized processing, Better maintainability, Reduced code duplication.

Interceptors in Struts provide reusable preprocessing and postprocessing services such as validation, exception handling, and logging, making web applications more modular and maintainable.

Q4c) Draw and Explain MVC Architecture for Developing Web Application

MVC Architecture - MVC stands for: Model - View - Controller

MVC is a software design pattern used to separate: Business logic, Presentation layer, Request handling. This separation improves maintainability and scalability of web applications.

Components of MVC -

Component Purpose

Model Handles business logic and database

View Displays data to user

Controller Receives requests and controls flow

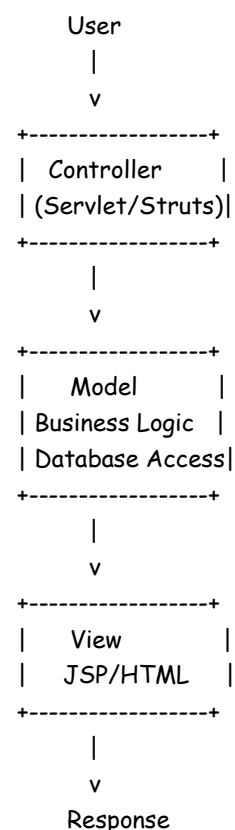
MVC Architecture Diagram -

Working of MVC Architecture

Step 1: User Request

User sends request through browser.

Step 2: Controller Processing



Controller receives request and decides appropriate action.

Example: Servlet, Struts Controller

Step 3: Interaction with Model

Controller communicates with Model to process: Business logic, Database operations

Step 4: Data Returned

Model returns processed data to Controller.

Step 5: View Generation

Controller forwards data to View (JSP).

Step 6: Response to User

View displays response to user.

Advantages of MVC Architecture - Separation of Concerns, Easy Maintenance, Reusability, Scalability, Parallel Development

Applications of MVC - Enterprise web applications, E-commerce websites, Banking systems

MVC architecture improves organization, maintainability, scalability, and reusability in web application development by separating application components into Model, View, and Controller layers.

Q4b) Write the benefits of Web services and explain SOAP, Rest and UDDIWeb

REST stands for: Representational State Transfer. REST is an architectural style used for developing lightweight web services.

REST uses HTTP methods such as: GET, POST, PUT, DELETE

Features of REST - Lightweight, Faster than SOAP, Uses JSON/XML, Stateless communication

REST Architecture -

Example of REST URL - `http://example.com/student/101`

Advantages of REST -

- Simple and fast,
- Less bandwidth usage
- Easy integration with mobile/web apps

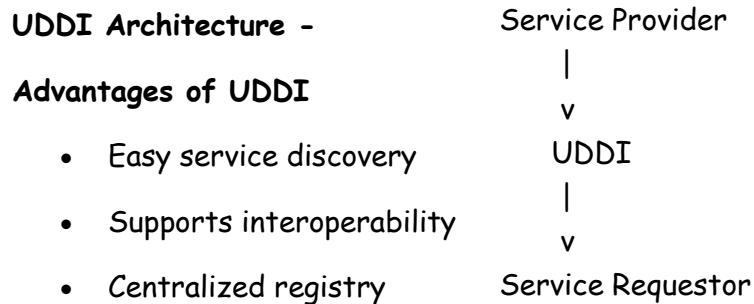


UDDI stands for: Universal Description Discovery and Integration

UDDI is a directory service used for publishing and discovering web services.

It acts like a registry for web services.

Functions of UDDI - Publish web services, Discover services, Describe services



Web services enable communication between heterogeneous applications. SOAP provides secure XML-based messaging, REST offers lightweight and faster communication, while UDDI helps publish and discover web services efficiently.